4 Discussion Week 2: Quadric Surfaces and Traces

Definition 4.1: Quadric Surface

A quadric surface is defined by a quadratic equation in three variables x, y, and z:

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + ax + by + cz + d = 0$$

, where $A, B, C, D, E, F, a, b, c, d$ are constants, and at least one of A, B, C, D, E, F is non-zero.

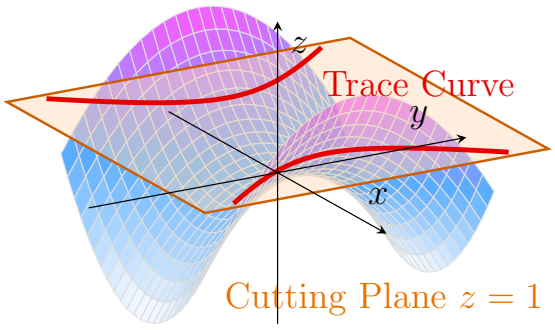
Definition 4.2: Trace

A trace is the intersection of a quadric surface with a plane, usually by taking slices parallel to the coordinate planes.

Note that when our surface can be written by $z = f(x, y)$, the traces form level sets of f.

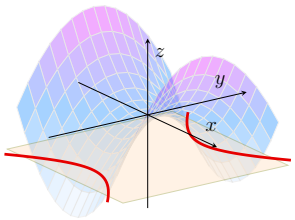
Example 4.1.

Consider the quadratic equation $z = x^2 - y^2$. Now take the traces in $z = -1$, $z = 0$, and $z = 1$. Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. In general, the traces in $x = k$ when k is a constant are hypobolas with equations $z = k^2 - y^2$.

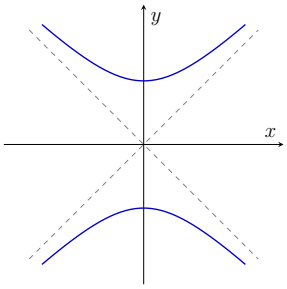


Now take the traces in $z = -1$, $z = 0$, and $z = 1$.

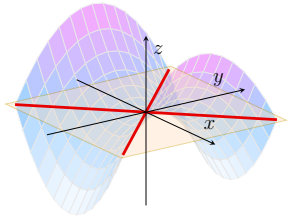
Trace at $z = -1$



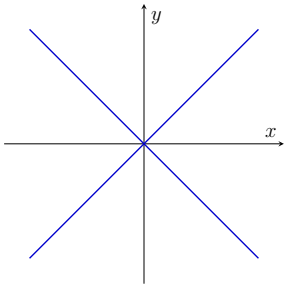
$y^2 - x^2 = 1$



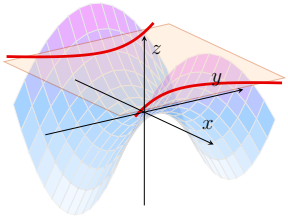
Trace at $z = 0$



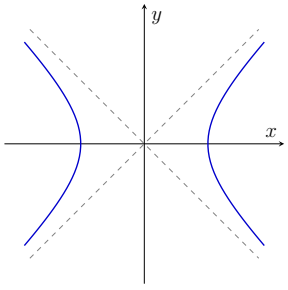
$y = \pm x$



Trace at $z = 1$

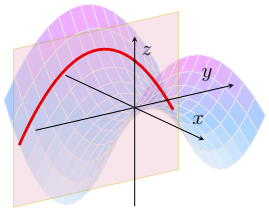


$x^2 - y^2 = 1$

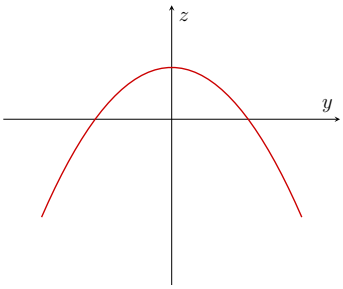


Make a sketch of the traces in $x = -1$, $x = 0$, and $x = 1$. Slicing the surface with vertical planes $x = k$ parallel to the yz-plane yields parabolic trace curves.

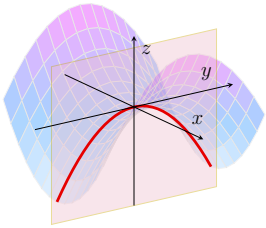
Trace at $x = -1$



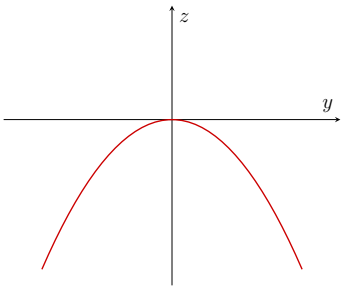
$z = 1 - y^2$



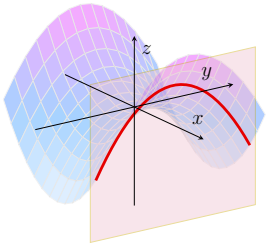
Trace at $x = 0$



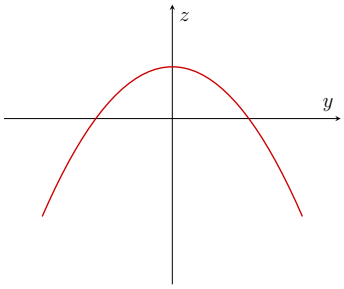
$z = -y^2$



Trace at $x = 1$



$z = 1 - y^2$



Ellipsoids are another family of quadric surfaces. The general equation of an ellipsoid is given by:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1$$